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Cloud Native Observability

Prometheus Architecture

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This article explores the architecture of Prometheus, focusing on its core components and functionality for monitoring time series data.

In this lesson, we explore the architecture of Prometheus, a powerful monitoring solution designed to collect, store, and query time series data. Prometheus is built around three core components: the data retrieval worker, the time series database, and the HTTP server.

Core Components of the Prometheus Server

Data Retrieval Worker

The data retrieval worker is responsible for gathering metrics from your targets. It does this by sending HTTP requests—typically to the `/metrics` endpoint—of your applications or systems. Once collected, these metrics are stored in the time series database, ready for analysis.

Time Series Database

The time series database serves as a dedicated repository for the collected metrics. Optimized specifically for time series data, it enables efficient recording and rapid retrieval of metric information.

HTTP Server

The HTTP server provides a query interface that allows users to access, visualize, and analyze stored metrics. By leveraging PromQL—Prometheus's built-in query language—

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Prometheus's functionality in dynamic environments.

Exporters and Data Collection

To scrape metrics effectively, Prometheus utilizes exporters. Exporters are lightweight processes running on your targets that expose metrics in a Prometheus-compatible format. Since many systems do not natively present metrics as expected by Prometheus, exporters are essential for converting internal data into a standardized format.

Prometheus employs a pull-based model, meaning it actively queries targets. However, for short-lived jobs that might not exist long enough to be scraped, the Pushgateway is used. This component temporarily stores metrics pushed by these jobs until Prometheus can scrape them.

Targets to scrape are usually specified in a static configuration file. In dynamic environments, such as Kubernetes or cloud infrastructures, service discovery mechanisms automatically update the target list.

Alerting

Prometheus supports alerting by evaluating collected metrics against defined thresholds. While it does not send notifications directly, it forwards alerts to Alertmanager. Alertmanager then manages and dispatches notifications through various channels like email, SMS, or Slack.

The overall interaction among these components is illustrated below:

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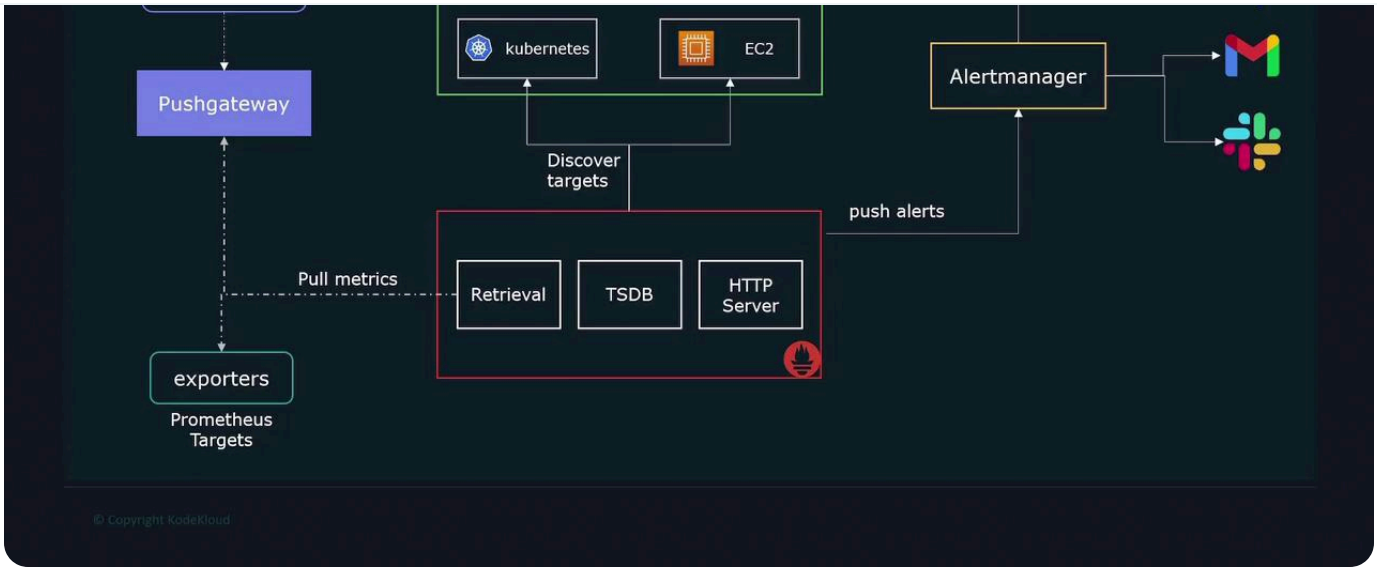

Querying Metrics with PromQL

Prometheus enables users to query and visualize metrics using PromQL, its powerful query language. Whether via the Prometheus web UI or third-party tools like Grafana, PromQL provides flexibility for retrieving and analyzing data. By default, Prometheus scrapes metrics from the `/metrics` endpoint of each target, though this endpoint can be customized in the configuration if needed.

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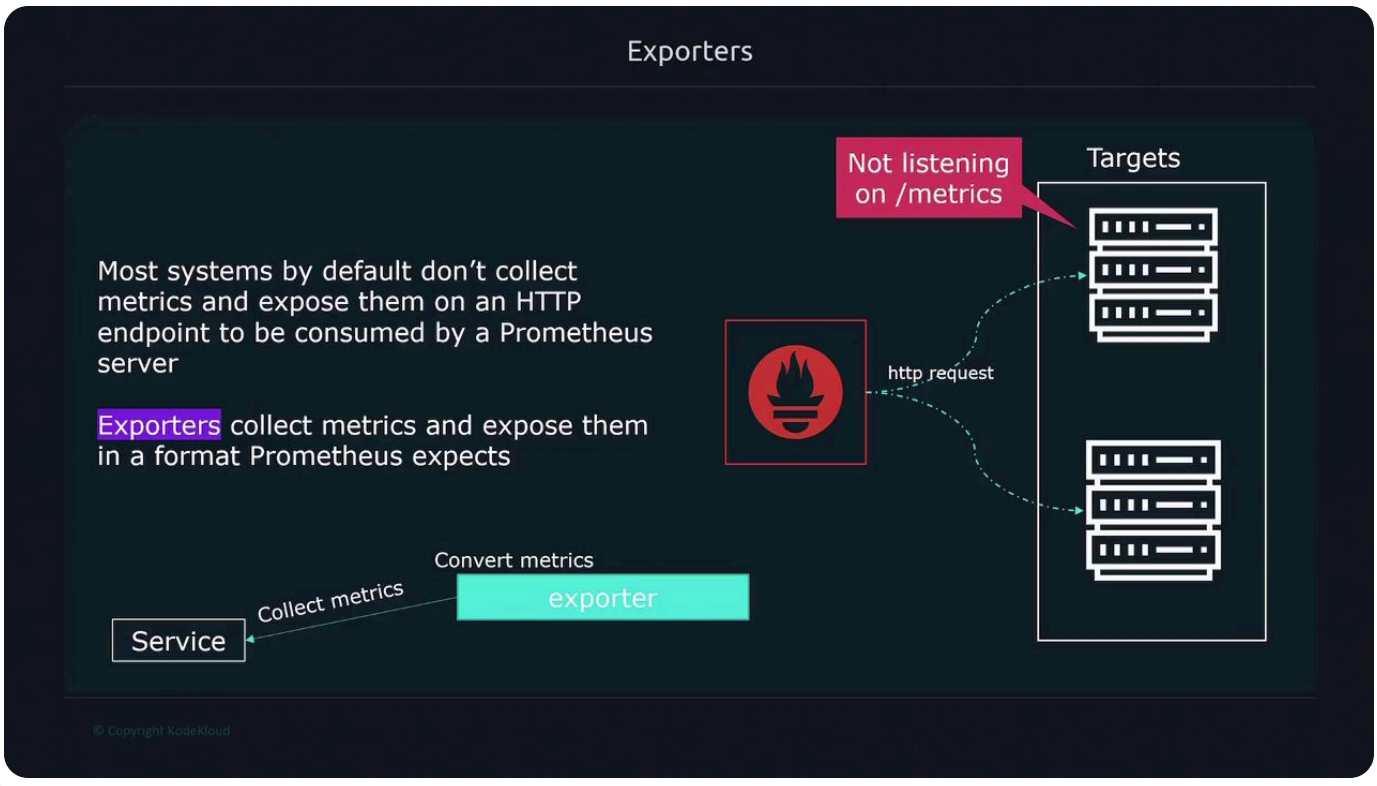
Many systems do not expose metrics in the required format; exporters bridge this gap by collecting metrics from applications, converting them into a compatible format, and exposing them on the `/metrics` endpoint for Prometheus to scrape.



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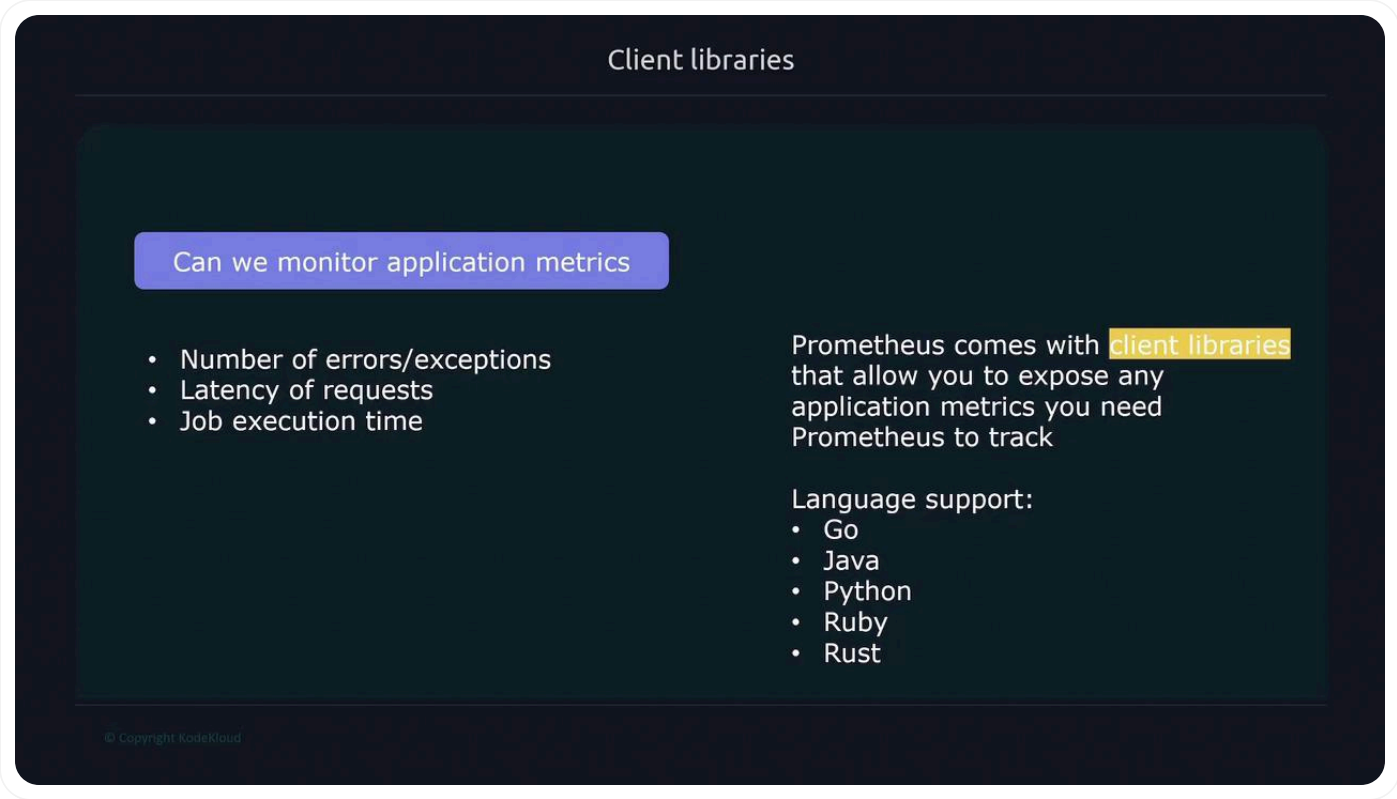
Custom Metrics Collection

For applications that require monitoring of custom metrics—such as tracking errors, latency, or execution time—Prometheus provides client libraries in multiple programming languages, including Go, Java, Python, Ruby, and Rust. These libraries enable you to expose application-specific metrics tailored to your needs.



Pull-Based vs. Push-Based Collection Models

Prometheus is primarily built around a pull-based model, meaning it scrapes metrics from known targets. This model offers several advantages:

Easier detection of targets that are down.

Control over server load, as metrics are collected at scheduled intervals.

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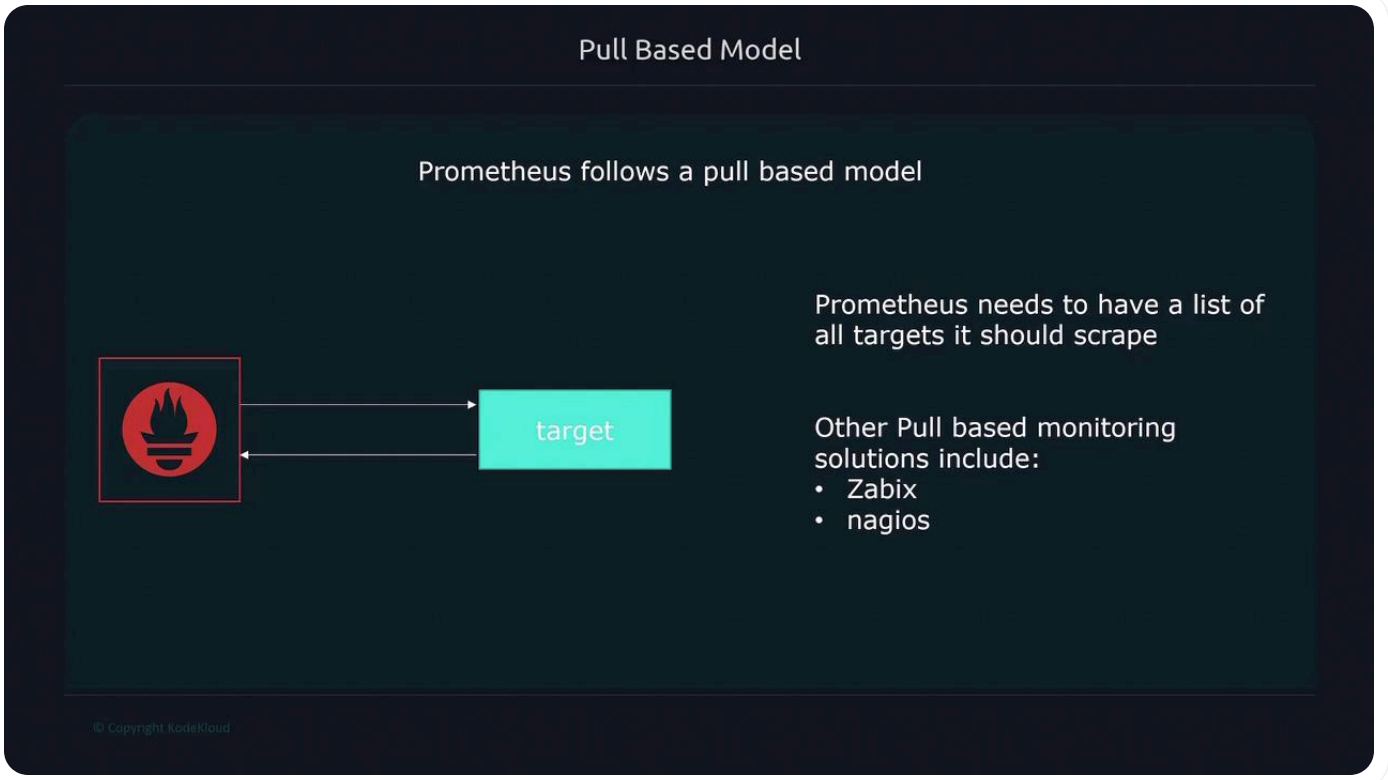
Pull Based Model
Prometheus follows a pull based model
target
Prometheus needs to have a list of all targets it should scrape
Other Pull based monitoring solutions include:
• Zabix
• nagios
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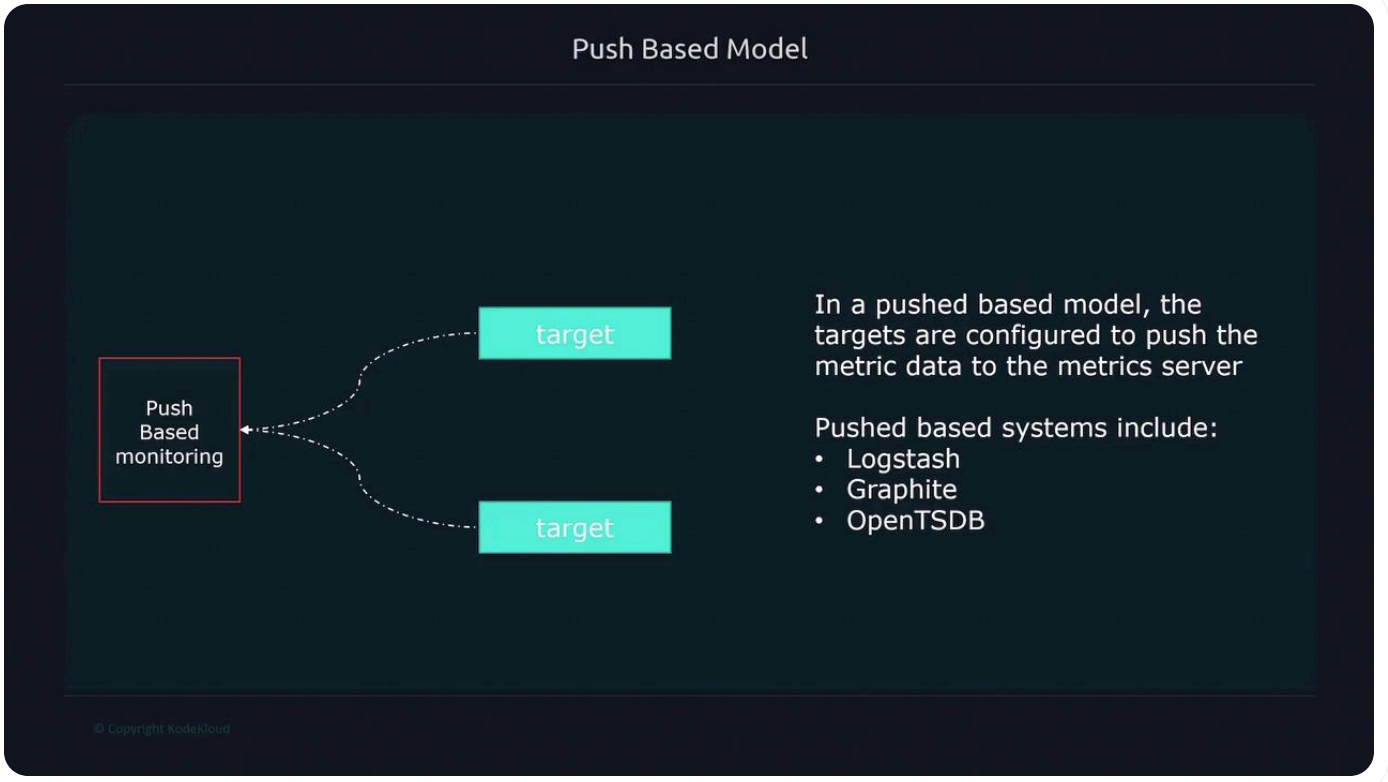
Push Based Model
Push Based monitoring
target
target
In a pushed based model, the targets are configured to push the metric data to the metrics server
Pushed based systems include:
• Logstash
• Graphite
• OpenTSDB
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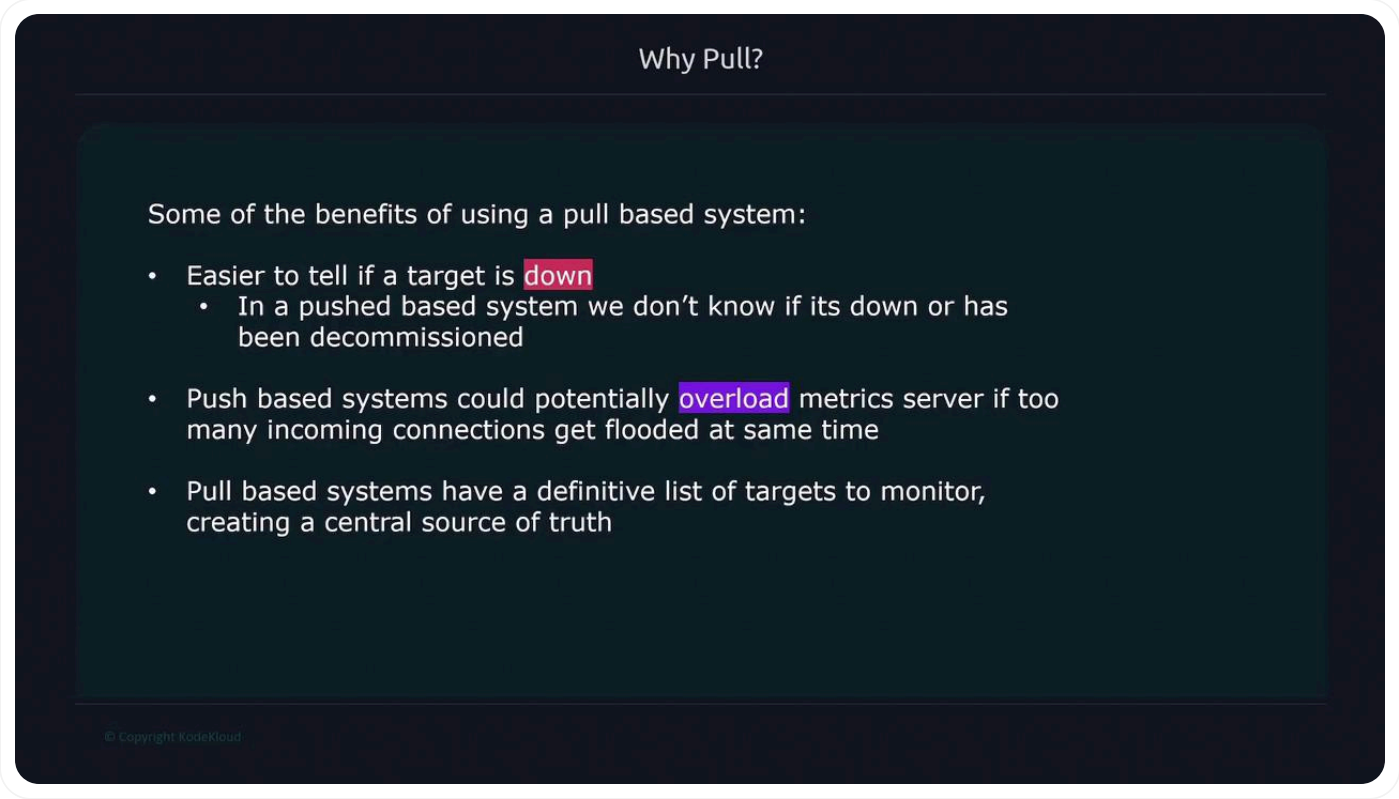
Reduced risk of server overload by controlling the rate of metric collection.

Maintaining a centralized registry of monitored targets.



Although the pull-based model is effective for numeric metrics, it may not be ideal for event-based data or short-lived jobs. In such cases, the Pushgateway allows these jobs to push their metrics for subsequent scraping by Prometheus.

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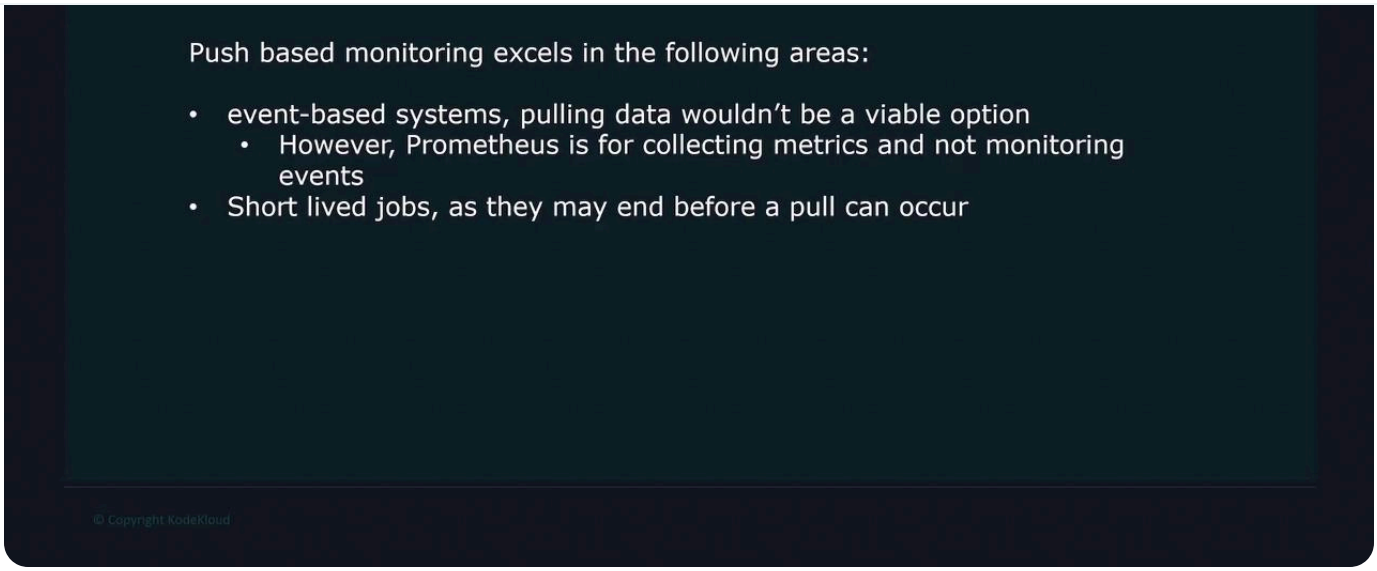

Conclusion

Prometheus is engineered for the efficient collection, storage, and querying of time series metrics. Its modular architecture—including key components like exporters, service discovery, and Alertmanager—ensures it can meet the monitoring requirements of both static and dynamic environments. By leveraging a pull-based model and providing push-based alternatives via Pushgateway, Prometheus offers a comprehensive monitoring solution for a wide variety of use cases.

For more detailed information, consult the **Prometheus Documentation**.

Learn about setting up exporters and configuring service discovery to streamline your monitoring setup.

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